

# Shaurya Varshney

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## EDUCATION

### Indian Institute of Science Education and Research (IISER)

Bhopal, India

Bachelor of Science in Engineering Sciences (Computer Science)

2023 – 2027 (Expected)

- **Academic Standing:** Maintained a current CPI of **7.31 / 10.0** within a rigorous, research-centric core scientific curriculum.
- **Core Coursework:** Machine Learning, Data Structures & Algorithms, Signals & Systems, Discrete Mathematics.

### Delhi Public School Ghaziabad (DPSG MRD)

Ghaziabad, India

Secondary & Higher Secondary Education

2019 – 2023

- Class XII: **90.2%** (Awarded Merit Scholarship for ranking in top academic tier) | Class X: **95.8%**

## RESEARCH & COMPUTATIONAL PROJECTS

### Object Detection and Multi-Modal Distance Estimation for ADAS | Python, OpenCV, YOLOv11

2025

- Developed a real-time Advanced Driver Assistance System (ADAS) leveraging fine-tuned YOLOv11 architectures, achieving a **94.2% mean Average Precision (mAP@0.5)** for multi-class object detection.
- Designed a hybrid mathematical localization framework fusing pinhole camera models and ground-plane geometry, reducing distance estimation error to **4.8%** across a 50-meter operational radius without stereoscopic hardware.
- Implemented multi-object state tracking optimized via Kalman Filtering to eliminate noise, reducing tracking identity switches (ID Sw) by **35%**.
- Constructed a low-latency safety zone classification matrix executing real-time spatial collision-risk telemetry at **60+ Frames Per Second (FPS)**.

### Stochastic Modeling & Quantitative Finance Pipelines | Python, Scikit-Learn, Pandas

2024

- Engineered predictive statistical models to map asset volatility trends by processing over **1.2M+ rows** of high-frequency historical market data.
- Utilized multi-variable regression and rolling-window statistical architectures to identify predictive signals, improving baseline backtest prediction accuracy by **18%**.

### Hardware Automation & Microcontroller Orchestration | C++, Embedded Systems

2024

- Designed and benchmarked low-level automated circuitry prototypes across **4 synchronous hardware nodes**, realizing a **15% optimization** in baseline sensory data ingestion latency via interrupt-driven execution paths.

## PROFESSIONAL EXPERIENCE

### HCLTech

Noida, India

Software Engineering Intern

May 2026 – August 2026

- Architected a production-grade, fault-tolerant log ingestion pipeline to process a high-throughput stream of **50M+ daily telemetry events**.
- Deployed **Vector** agents across distributed microservice topologies, reducing edge node CPU and memory utilization by **40%** compared to legacy Logstash configurations.
- Developed high-concurrency stream-processing edge services using **Node.js** to validate schemas at a throughput of **12,000+ requests per second**.
- Containerized operations via **Docker** and engineered optimized data sinks into **ClickHouse**, reducing historical storage footprint by **65%** and cutting analytical query latency to **150ms**.

### Transorg Analytics

Gurugram, India

Generative AI Intern

May 2025 – July 2025

- Built an end-to-end proprietary Resume Parser and interactive knowledge graph execution engine utilizing Retrieval-Augmented Generation (RAG) to parse **5,000+ unstructured technical profiles**.
- Achieved a **27% improvement** in semantic document retrieval accuracy (Hit Rate@5) via custom embedding fine-tuning, recursive parent-child text chunking, and algorithmic prompt engineering.
- Exposed secure, asynchronous microservices via FastAPI backend layers, reducing model token generation latency by **30%** through hardware-optimized local Ollama orchestration.

## LEADERSHIP & SOCIAL IMPACT

### Green Lungs Universal Foundation (GLUF) | Co-Founder

2025 – Present

- Co-founded an impact-driven environmental foundation, scaling operations across **3 municipal jurisdictions** and coordinating a volunteer network of **120+ active members**.
- Directed technical and operational roadmaps to implement a digital campaign ledger, optimizing regional resource allocation efficiency by **22%** across all active tracking nodes.

## TECHNICAL SKILLS

**Programming Languages:** Python, C, C++, SQL, LaTeX, MATLAB, JavaScript (ES6+)

**AI, ML, & Frameworks:** LangChain, RAG, Ollama, FAISS, OpenCV, YOLOv11, Scikit-Learn, NumPy, Pandas, Matplotlib

**Backend & Databases:** FastAPI, Node.js, PostgreSQL, ClickHouse, Neo4j, Telemetry Systems

**Tools & Infrastructure:** Git, GitHub, Docker, Vector, Unix/Linux Environments, VS Code

## ACTIVITIES & ACHIEVEMENTS

- **House Captain:** Elected to govern an academic cohort of **40+ students**; structured **4 competitive developmental symposia** to increase engagement.
- **Media Lead:** Supervised targeted multi-channel digital outreach campaigns, scaling organizational audience registration and attendance metrics by **25%**.